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Binary Komodo Mlipir Algorithm for Feature Selection

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***Abstract*—Komodo Mlipir Algorithm (KMA) is a metaheuristic algorithm that mimics the behavioral patterns of Komodo and the unique walking manner called "mlipir" in the Javanese language. Nevertheless, the KMA is a recently developed algorithm that has not yet been methodically applied for feature selection issues. This work presents a variant of the Komodo Mlipir Algorithm (BKMA) that is specifically tailored to tackle binary issues, with a particular focus on feature selection. The proposed approach consists of eight alternative transfer function variations, denoted by the forms S and V, with each shape having four distinct versions. The BKMA is composed of two binarization rules, namely Standard and Elitist, which are combined to create a total of sixteen versions. The comparison is performed by transforming a continually search area into a discrete search area. In order to assess the efficacy of the proposed approach, a total of 12 benchmark datasets were collected from the repository curated by the University of California, Irvine. The K-nearest neighbors (KNN) algorithm is used inside an objective function to facilitate the selection of the suggested features. The study's results suggest that the suggested strategy outperforms others in terms of both the fitness value and the classification accuracy.**

***Index Terms*—Binarization schemes, binary optimization, feature selection, metaheuristics, transfer functions**

I. Introduction

Information technology is used in economics, medical research, engineering, agriculture, technology, and bioinformatics due to its rapid progress. The curse of dimensionality has increased the time and complexity of computer processes due to the expansion of features [1]–[3]. Complex data with redundant and unimportant attributes hinders machine learning and accuracy. One must establish a rational plan to solve this problem. Most study domains rely on feature selection to carefully examine and choose relevant characteristics to solve problems. A reasonable method is needed to solve this challenge.

Choosing relevant and non-duplicative features from a dataset is called feature selection [4]. Although feature selection is computationally challenging (NP-hard), it does not guarantee the best answer [5]. This is a discrete issue, especially binary optimization [6]. Feature selection has a complexity of $2^n - 1$ or $O(2^n)$, where n is the total number of accessible features [5], [7]. Liu and Motoda [8] classify feature selection methods by subset assessment and search procedure. Regarding the first, feature selection is often divided into three methods: filter, wrapper, and embedded [9].

The Komodo Mlipir Algorithm is a metaheuristic algorithm developed by Suyanto et al. [10], gaining inspiration from the hunting behavior of Komodo dragons. Similar to other metaheuristic algorithms, KMA operates inside a continuous search space, making it suboptimal for solving the binary feature selection issue. Hence, it is necessary to implement a transition from KMA in order to effectively tackle the current situation. The research will involve the continuous transformation of a search space into a discrete or binary representation using a Two-Step approach that employs a transfer function [11]. A binary version of KMA will use the Wrapper method, which uses KNN as the classification algorithm.

The contribution of this research is as follows:

- We propose BKMA, binary variant of KMA. As far as we know, this is the initial instance where KMA is being employed to address the issues by utilizing a binary search space.
- Apply a binary search space to the implementation of KMA for feature selection in order to tackle the issue of curse of dimensionality.
- Data classification was performed using 12 distinct benchmark datasets collected from the UCI repository.

The rest of this paper is split as follows: In Section II, the fundamentals of KMA and suggested binary forms for feature selection are outlined. See Section III for information on the proposed Binary KMA. In Section IV, the experiment setup, results, and findings are presented. Finally, Section V discusses conclusions and future directions.

II. Methods

A. Komodo Mlipir Algorithm (KMA)

In 2022, Suyanto et al. developed the Komodo Mlipir Algorithm (KMA), a population-wide probabilistic approach inspired by the behavior of Komodo dragons from East Nusa Tenggara and the Javanese concept of "mlipir" [10]. KMA categorizes Komodo dragons into three groups for optimal problem-solving: Big Adult Komodos, Female Komodos, and Small Males. The population p of n Komodos is divided into b big and s small males using the following formulas:

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$$b = \lfloor (p-1)n \rfloor \tag{1}$$

$$s = n - b \tag{2}$$

Big Adult Komodos hunt large prey and may share their meals with smaller Komodos. The movement of big males t_i to a new position $t_i^{'}$ is defined as:

$$w_{ij} = \begin{cases} r_1(t_j - t_i) & \text{if } f(t_j) < f(t_i) \text{ or } r_2 < 0.5 \\ r_1(t_i - t_j) & otherwise \end{cases} \tag{3}$$

$$t_i^{'} = t_i + \sum_{b}^{j=1} w_{ij}, \text{ where } j \neq i \tag{4}$$

Female Komodos may reproduce sexually or via parthenogenesis. In sexual reproduction, the offspring positions are given by:

$$\begin{aligned} t_{il}^{'} &= (1 - r_l) \cdot t_{jl} + r_l \cdot t_{il} \\ t_{jl}^{'} &= (1 - r_l) \cdot t_{il} + r_l \cdot t_{jl} \end{aligned} \tag{5}$$

In parthenogenesis, new values are generated stochastically using a normal distribution:

$$(t_{i1}, t_{i2}, ..., t_{im}) \rightarrow (t_{i1}^{'}, t_{i2}^{'}, ..., t_{im}^{'}) \tag{6}$$

$$t_{ij}^{'} = t_{ij} + (2r - 1)\alpha \, |ub_j - lb_j| \tag{7}$$

Vector element $t_{i1}, t_{i2}, ..., t_{im} \in [lb_j, ub_j]$ reflects the m-dimensional features of an individual **k** in the Komodo population. The values lb_j and ub_j are the lower and upper limits of the j_{th} dimension. A random number r drawn from a normal distribution and a predetermined radius α of 0.1 allow for new solutions within 10% of the search space.

Small males avoid big males and seize food opportunistically. The $i-th$ Komodo follows the $j-th$ one at a constant mlipir rate of d, where d is a value between 0 and 1.

$$w_{ij} = \begin{cases} \sum_{l=1}^{m} r_1(t_{jl} - t_{il}) & \text{if } r_2 < d \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

$$t_i^{'} = t_i + \sum_{j=1}^{b} w_{ij}, \text{ where } j \neq i \tag{9}$$

KMA uses three variables: population size n, portion p, and mutation rate d. p and d are typically set to 0.5, but $n^{'}$ is adapted during evolution based on fitness improvements:

$$n^{'} = \begin{cases} n - 1 & \text{if } \delta f_1 > 0 \text{ and } \delta f_2 > 0 \\ n + 1 & \text{if } \delta f_1 = 0 \text{ and } \delta f_2 = 0 \end{cases} \tag{10}$$

KMA operates as follows: After initializing n random Komodo dragons, the algorithm evolves until a stopping condition is met. The population is ranked, with b big males, one female, and s small males, as defined by Equations (1) and (2). Big males are moved per Equation (4), females reproduce per Equations (5) and (7), and small males are moved per Equation (9). Population size is dynamically adjusted using Equation (10). The global optimum is the Komodo k_{best} with the highest quality after meeting the stopping criteria.

B. Feature Selection Problem

The Feature Selection (FS) problem is a combinatorial optimization issue arising from the need to eliminate unnecessary and redundant information, which hampers the effectiveness of prediction or classification algorithms [1]. The primary objectives are to improve the accuracy of classification algorithms and to reduce the number of features in the dataset.

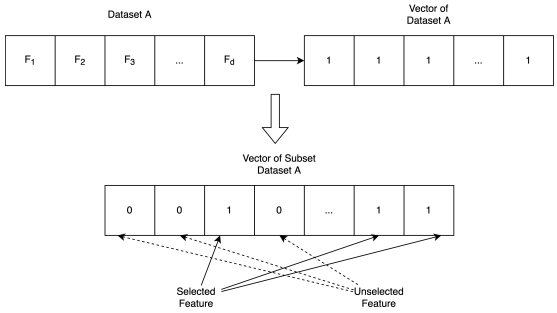


Fig. 1. Binary representation for feature selection.

FS operates on a dataset A with F features, where F is a positive integer, so that $S = F_1, F_2, F_3, ..., F_d$. The goal is to select the optimal subset $K = f_1, f_2, f_3, ..., f_n$ where $n < m$ ensuring the chosen features best represent the original dataset.

In the binary domain of FS, both the vector of Dataset A and the subset Dataset A have the same nature, as shown in Figure 1. Here, a value of 1 indicates the inclusion of a feature, while a value of 0 signifies its exclusion. The subset vector represents the selected features with 1s and 0s.

C. Applying Continuous Metaheuristics for Solving Binary Problems Using Two-Step

Binarization methods in continuous multi-objective metaheuristics convert continuous domain values into binary domains to maintain operational effectiveness and produce high-quality binary solutions. Research shows that continuous metaheuristics with a binary scheme outperform those without one on NP-hard combinatorial problems [12].

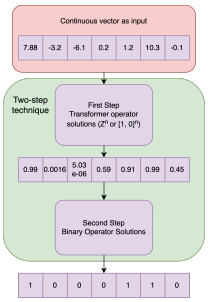


Fig. 2. Scheme two-step binarization techniques.

Two main types of binary scheme are used to solve combinatorial problems [12]. The first includes operators that do not alter metaheuristic activities, such as two-step methods, encoding transformation, and angle modulation. The second type is binary schemes that affect metaheuristic behavior,

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like the quantum binary technique [13], affects metaheuristic behavior by using quantum mechanics.

Recently, two-step techniques have become the most popular due to their superior convergence, ability to find optimal solutions, and ease of implementation [14]. This method involves first converting continuous responses and then transforming them into binary form (0 or 1) [15]. See Figure 2 for a diagram of the two-step process. In the first phase, a transfer function converts continuous MH values into a 0-1 range. In the second phase, these values are discretized into binary form, efficiently solving combinatorial problems and yielding high-quality binary solutions [16].

1) Transfer Functions: The transfer function is an efficient operator that generates probability values and aims to simulate the movement of individual locations. This function is responsible for the initial stage of the binarization process and converting R^n solutions into solutions within the range of $[0, 1]^n$.

TABLE I
S-SHAPED AND V-SHAPED TRANSFER FUNCTION

S-shaped		V-shaped	
Name	Transfer Function	Name	Transfer Function
S1	$T(d_w^j) = \frac{1}{1+e^{-2d_w^j}}$	V1	$T(d_w^j) = \left\lvert erf\left(\frac{\sqrt{\pi}}{2} d_w^j\right)\right\rvert$
S2	$T(d_w^j) = \frac{1}{1+e^{-d_w^j}}$	V2	$T(d_w^j) = \lvert\tanh(d_w^j)\rvert$
S3	$T(d_w^j) = \frac{1}{1+e^{\frac{-d_w^j}{2}}}$	V3	$T(d_w^j) = \left\lvert \frac{d_w^j}{\sqrt{1+(d_w^j)^2}}\right\rvert$
S4	$T(d_w^j) = \frac{1}{1+e^{\frac{-d_w^j}{3}}}$	V4	$T(d_w^j) = \lvert \frac{2}{\pi} \arctan\left(\frac{\pi}{2} d_w^j\right)\rvert$

Transfer functions include S, V, O, Z, X, Q, and U. The transfer functions S and V are essential for basic functions and are often utilized in literature [16]. Four ways each is used across parts. This study will use and define all elements described in this article in Table I.

TABLE II
BINARIZATION RULES

Type	Binarization Rules
Standard	$X_{new}^j = \begin{cases} 1 & \textit{if rand} \le T(d_w^j) \\ 0 & \textit{else} \end{cases}$
Complement	$X_{new}^j = \begin{cases} \textit{Complement}(X_w^j) & \textit{if rand} \le T(d_w^j) \\ 0 & \textit{else} \end{cases}$
Static Probability	$X_{new}^j = \begin{cases} 0 & \textit{if rand} \le T(d_w^j) \\ X_w^j & \textit{if } \alpha < T(d_w^j) \le \frac{1}{2}(1+\alpha) \\ 1 & \text{if } T(d_w^j) \ge \frac{1}{2}(1+\alpha) \end{cases}$
Elitist	$X_{new}^j = \begin{cases} X_{best}^j & \textit{if rand} < T(d_w^j) \\ 0 & \textit{else} \end{cases}$
Roulette Elitist	$X_{new}^j = \begin{cases} P[X_{new}^j = \zeta_j] = \frac{f(\zeta)}{\sum_{\delta \in Q_G}} & \textit{if rand} \le T(d_w^j) \\ P[X_{new}^j = 0] = 1 & \textit{else} \end{cases}$

2) Binarization Rules: The two-step process uses a binarization procedure to turn the position into a binary answer of 0 or 1. According to [15], J. Lemus-Romani et al. established five ways for binarization rule. All techniques are listed in Table II. Equation Standard is most often used in the literature, although complement, static probability, elitism, and roulette are all employed.

III. PROPOSED METHOD BINARY KOMODO MLIPIR ALGORITHM (BKMA)

A. Proposed Approaches

Feature selection is a binary optimization problem [1]. Each feature from the original feature vector can be included or excluded, represented as a binary vector of 0s and 1s. A 0 indicates exclusion, while a 1 indicates inclusion (see Figure 1). The Komodo vector population is represented by this binary vector.

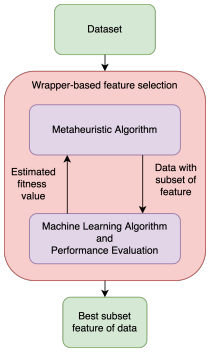


Fig. 3. Wrapper Feature Selection process.

The proposed binary KMA employs a two-step approach, as detailed in Section II-C. In the original KMA [10], Komodos navigate the continuous real domain. In the binary search space, position updates switch between 0 and 1. This is achieved using a two-step approach involving a transfer function and binarization rule.

The transfer function determines the probability of changing binary values from 0 to 1 or vice versa, guiding Komodo's movement in binary space. This study uses all variations of S-shaped and V-shaped transfer functions, converting Komodo's velocity d_w^j into probabilities ranging from 0 to 1. Table I displays the formulas for the eight transfer functions utilized.

The binarization rules used to transform positions ranging from 0 to 1 into binary solutions for this study are the Standard and Elitist, as described mathematically on Table II. The complete process is equivalent to the diagram shown in Figure 3.

B. Fitness Function

Assessing the selected feature subset is crucial in feature selection (FS) techniques. In this study, a wrapper-based approach using the KNN classifier was employed.

The K-Nearest Neighbors (KNN) classifier is the favored choice for binary metaheuristics in feature selection because of its simplicity, efficacy on small datasets, non-parametric nature, and ability to adapt to different distance measurements. KNN maintains the local data structure, which is essential for finding pertinent characteristics, and effectively manages binary data without requiring intricate conversions. It seamlessly incorporates with metaheuristic algorithms, enabling the fitness function to efficiently analyze feature subsets. Although other classifiers are available, the combination of simplicity,

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versatility, and efficiency in KNN makes it an appealing option for this particular application.

The primary goals of FS include limiting the number of features and enhancing classification accuracy by removing redundant and irrelevant attributes. Balancing these goals is essential since higher accuracy often conflicts with fewer features. To address this, the classification error rate $E_R(D)$ of the KNN classifier is used as a fitness function (Equation 11):

$$fitness = \alpha E_R(D) + \beta \frac{|S|}{|T|} \tag{11}$$

Here, $|S|$ denotes the number of selected features, and $|T|$ denotes the total number of features in the dataset. The parameters α are values between 0 and 1 and β (where $\beta = (1 - \alpha)$) balance the importance between classification error rate and subset size.

IV. Experimental Results and Analysis

A. Dataset

The suggested binary BKMA is verified using 12 benchmark data sets from the UCI repository [17]. The table III displays details about the specified data sets, such as number of instances, features, and classes. The table illustrates that the method was tested in low, medium, and large search regions using data sets of different instance and feature sizes. These regions show the amount of features per dimension and number of instances for the input size.

TABLE III
Overview of the datasets utilized in the current study

No	Dataset	No. of features	No. of instances	No. of classes
1	Breastcancer	9	699	2
2	BreastEW	30	568	2
3	Exactly	13	1000	2
4	HeartEW	13	270	2
5	Ionosphere	34	351	2
6	KrVsKpEW	36	3196	2
7	PenglungEW	325	73	2
8	Sonar	60	208	2
9	Vote	16	300	2
10	Wine	13	178	3
11	Zoo	16	101	6
12	Toxicity	1203	171	2

B. Experimental Setup and Parameter Settings

This experiment was done on an M1 MacBook Pro with 8 GB of RAM using Python 3.11.7. Table IV displays algorithm-specific parameter settings. The algorithms running 100 repetitive experiments, repeated 15 times, to get an average performance.

C. Evaluation Metrics

The data for validation, training, and testing is separated into three equal pieces using a random process. The process of segregating the data is performed multiple times in order to guarantee robust and quantifiable outcomes. During each iteration, the validation data is utilized to evaluate the following statistical metrics:

TABLE IV
Parameter Configuratoin

Parameter	Value
Population size	10
Maximal no. of iteration metaheuristic	100
Independent runs	15
Dimension	No. of features
Search domain ([ub, lb])	[0,1]
Parameter k of KNN	5
Metrics of KNN	minkowski
p value of KNN	2
α in fitness function	0.99

1) Average of Classification Accuracy:

$$Avg\ Accuracy = \frac{1}{N} \sum_{k=1}^{N} Accuracy^k \tag{12}$$

where $Accuracy^k$ is the accuracy achieved at run k.

2) Mean of Fitness:

$$Avg\ Fitness = \frac{1}{N} \sum_{k=1}^{N} Fitness^k \tag{13}$$

where $Fitness^k$ is avarage fitness achieved during run k.

D. Assessment of The Proposed BKMA Variants

The average classification accuracy and standard deviations of BKMA are shown in Tables V and VI, Elitist binarization trumps Standard binarization in both tables. In BKMA, Elitist binarization had 6 variations that ranked highest, whereas Standard binarization had 4. The fitness value of Elitist binarization was lower, with 5 variations placing first compared to 3 for Standard binarization.

BKMA-SS3 outperformed Standard binarization BKMA algorithms on 2 of 12 datasets in classification accuracy. Further, BKMA-SS1, BKMA-SS2, and BKMA-SS4 performed best on 1 of 12 datasets. However, Elitist binarization rules produced competitive results. Table VI shows that BKMA-ES1 and BKMA-ES3 often score top among BKMA Elitist versions. BKMA-ES1 scored 5 out of 12 datasets, whereas BKMA-ES3 scored 3. The BKMA-ES3 model achieved 100% accuracy on the Wine and Zoo datasets. Although BKMA-EV1 performed poorly, it had a much better accuracy of 82% in the Toxicity dataset, which includes 1203 features that mean has a dimensionality of 1203 for computation. All BKMA variants surpass classification without feature selection.

Tables VII and VIII show mean fitness of all BKMA variations The table VII shows that BKMA-SS1 achieves the finest degree of fitness in 3 of 12 datasets. BKMA-SS1 outperforms all other BKMA variants in fitness. BKMA-ES4 achieves many lowest fitness than other elitist binarization, that got 2 first rank out of 12 datasets. BKMA-ES1, BKMA-ES2, and BKMA-ES3 obtained the same rank first sum of 1 from 12 datasets. The BKMA-EV1 algorithm handles large Toxicity datasets well, resulting in the lowest fitness value among all BKMA versions. Elitist varieties exceed Standard in fitness, with 5 of 8 ranking top.

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Mathematical aspects may explain transfer functions' influence. S-shaped transfer functions or Sigmoid functions provide smoother transitions or gradual shifts, which may help prevent local optima. Functions with a V-shaped curve may cause more aggressive changes that avoid local optima but may cause instability. These theoretical ideas are supported by our empirical findings. BKMA-SS1 and BKMA-SS3 used S-shaped transfer functions and had good classification accuracy and stable performance. However, BKMA versions with V-shaped transfer functions, such BKMA-EV1, performed well on high-dimensional datasets like the Toxicity dataset, where quick and decisive transitions were favorable.

The Elitist approach outperforms Standard binarization rules due to considerable differences. Elitism is used in the Elitist binarization algorithm to choose high-quality features that performed well in previous rounds. This method preserves and prioritizes the best answers, making search space exploration and exploitation more efficient. However, the Standard binarization strategy treats all attributes equally regardless of significance or quality. This may result in a less concentrated, random search. Multiple datasets show that Elitist binarization has higher classification accuracy and lower fitness values. BKMA-EV1's performance on the Toxicity dataset, which comprises 1203 features, showed that the Elitist approach can handle high-dimensional data. Theoretical rationales for the superiority of Elitist binarization may be found in its ability to exert selective pressure, which helps in maintaining high-quality solutions and directing the search process towards more promising parts of the search space. This selective pressure is especially advantageous in preventing suboptimal solutions and ensuring a more resilient exploration process. Conversely, the Standard binarization approach without selectivity may provide weaker feature subsets and lower performance.

Based on both fitness and classification accuracy, it can be determined that the variance BKMA-SS1, which represents the standard binarization rules, and BKMA-ES1, which represents the elitist binarization, are superior to the other variants.

V. Conclusion and Future Direction

Feature selection (FS) is crucial in Data Mining and Machine Learning. This study introduces a binary variant of the Komodo Mlipir Algorithm (BKMA) tailored for feature selection. Originally designed for continuous search spaces, BKMA has been adapted to binary search spaces using S-shaped and V-shaped transfer functions, alongside Standard and Elitist binarization methods. Sixteen BKMA approaches were tested on twelve standard UCI datasets to identify optimal feature subsets, emphasizing the impact of transfer functions on classification accuracy.

When designing the fitness function, we have considered both the precision of classification and the number of selected characteristics. In order to assess the efficacy of the proposed techniques, we have examined the most superior iteration of BKMA among the sixteen possible variations. The results suggest that BKMA surpasses KNN classification algorithms in terms of classification accuracy. Both in terms of fitness and classification accuracy, it is evident that the variance BKMA-SS1, which represents the standard binarization rules, and BKMA-ES1, which represents the elitist binarization, outperform the other variations.

Our future plans include attempting to combine BKMA with other metaheuristic FS techniques through hybridization. Furthermore, since the proposed FS version of BKMA is a universal method, it can easily be used in other engineering problem that often include high-dimensional vectors. Additionally, it is necessary to do a comparative analysis with various metaheuristic algorithms.

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TABLE V
AVERAGE CLASSIFICATION ACCURACIES OF THE PROPOSED FS METHOD USING EIGHT DIFFERENT TRANSFER FUNCTIONS AND STANDARD BINARIZATION RULE

	Without FS		BKMA-SS1		BKMA-SS2		BKMA-SS3		BKMA-SS4		BKMA-SV1		BKMA-SV2		BKMA-SV3		BKMA-SV4	
	Avg Acc	Std Acc	Avg Acc	Std Acc	Avg Acc	Std Acc	Avg Acc	Std Acc	Avg Acc	Std Acc	Avg Acc	Std Acc	Avg Acc	Std Acc	Avg Acc	Std Acc	Avg Acc	Std Acc
Breastcancer	0,609	0,0171	0,982	0,0099	0,979	0,0103	0,984	0,0124	**0,986**	**0,0092**	0,984	0,0117	0,982	0,0034	0,976	0,0179	0,981	0,0142
BreastEW	0,928	0,0216	0,960	0,0044	0,970	0,0142	**0,979**	**0,0103**	0,968	0,0120	0,958	0,0088	0,958	0,0168	0,968	0,0162	0,951	0,0118
Exactly	0,725	0,0404	0,980	0,0032	**0,991**	**0,0073**	0,971	0,0395	0,925	0,0470	0,825	0,0365	0,821	0,0213	0,867	0,0573	0,823	0,0890
HeartEW	0,663	0,0427	0,889	0,0117	0,870	0,0332	**0,896**	**0,0189**	0,889	0,0262	0,855	0,0516	0,855	0,0361	0,870	0,0165	0,837	0,0139
Ionosphere	0,851	0,0350	0,932	0,0329	0,910	0,0278	0,933	0,0140	0,904	0,0340	0,918	0,0190	0,912	0,0108	0,941	0,0244	0,904	0,0105
KrVsKpEW	0,960	0,0067	0,972	0,0031	0,971	0,0052	0,974	0,0037	0,972	0,0031	0,973	0,0050	0,968	0,0052	0,968	0,0081	0,969	0,0117
PenglungEW	0,760	0,0905	0,893	0,0532	0,920	0,0497	0,893	0,0679	0,933	0,0210	0,947	0,0652	0,960	0,0533	0,960	0,0533	0,907	0,0532
Sonar	0,781	0,0844	0,924	0,0176	0,910	0,0410	0,905	0,0259	0,919	0,0491	0,891	0,0390	0,900	0,0410	0,890	0,0441	0,900	0,0180
Vote	0,883	0,0281	**0,997**	**0,0068**	0,980	0,0194	0,970	0,0194	0,987	0,0124	0,970	0,0123	0,973	0,0132	0,967	0,0104	0,957	0,0136
Wine	0,722	0,0307	0,989	0,0137	0,983	0,0224	0,989	0,0137	0,989	0,0224	0,972	0,0177	0,967	0,0271	0,955	0,0137	0,972	0,0177
Zoo	0,848	0,0188	0,990	0,0192	0,990	0,0192	0,971	0,0235	0,981	0,0235	0,981	0,0235	0,981	0,0380	0,982	0,0356	0,981	0,0235
Toxicity	0,623	0,0610	0,703	0,0389	0,714	0,0478	0,749	0,0112	0,720	0,0553	0,754	0,0291	0,777	0,0214	0,789	0,0293	0,777	0,0214
average	0,7793		0,9342		0,9324		0,9345		0,9310		0,9190		0,9212		0,9277		0,9132	
rank first	0		1		1		2		1		0		0		0		0	

TABLE VI
AVERAGE CLASSIFICATION ACCURACIES OF THE PROPOSED FS METHOD USING EIGHT DIFFERENT TRANSFER FUNCTIONS AND ELITIST BINARIZATION RULE

	BKMA-ES1		BKMA-ES2		BKMA-ES3		BKMA-ES4		BKMA-EV1		BKMA-EV2		BKMA-EV3		BKMA-EV4	
	Avg Acc	Std Acc	Avg Acc	Std Acc	Avg Acc	Std Acc	Avg Acc	Std Acc	Avg Acc	Std Acc	Avg Acc	Std Acc	Avg Acc	Std Acc	Avg Acc	Std Acc
Breastcancer	**0,986**	**0,0044**	0,985	0,0052	0,980	0,0086	0,983	0,0117	0,977	0,0107	0,972	0,0112	0,970	0,0139	0,982	0,0034
BreastEW	0,965	0,0124	0,972	0,0036	0,973	0,0136	0,968	0,0154	0,968	0,0105	0,947	0,0054	0,953	0,0182	0,960	0,0072
Exactly	0,952	0,0393	0,934	0,0515	0,951	0,0418	0,967	0,0360	0,813	0,0218	0,821	0,0462	0,831	0,0250	0,785	0,0345
HeartEW	**0,896**	**0,0277**	0,874	0,0320	0,885	0,0214	0,892	0,0319	0,856	0,0412	0,852	0,0387	0,866	0,0445	0,852	0,0422
Ionosphere	**0,955**	**0,0224**	0,952	0,0143	0,952	0,0168	0,949	0,0248	0,932	0,0142	0,904	0,0276	0,915	0,0157	0,921	0,0073
KrVsKpEW	**0,977**	**0,0024**	0,970	0,0042	0,971	0,0067	0,972	0,0054	0,972	0,0046	0,971	0,0042	0,964	0,0027	0,968	0,0030
PenglungEW	0,933	0,0421	0,946	0,0268	**0,987**	**0,0268**	0,973	0,0328	0,906	0,0532	0,907	0,0799	0,933	0,0421	0,973	0,0328
Sonar	**0,938**	**0,0113**	0,929	0,0210	0,929	0,0210	**0,938**	**0,0240**	0,900	0,0235	0,876	0,0180	0,881	0,0402	0,910	0,0180
Vote	0,993	0,0083	0,980	0,0064	0,990	0,0083	0,990	0,0083	0,957	0,0341	0,953	0,0127	0,963	0,0164	0,970	0,0123
Wine	0,994	0,0112	0,983	0,0137	**1,000**	**0,0000**	0,994	0,0112	0,961	0,0282	0,972	0,0011	0,966	0,0112	0,994	0,0274
Zoo	0,990	0,0192	0,990	0,0192	**1,000**	**0,0000**	0,990	0,0192	0,990	0,0192	0,971	0,0381	0,962	0,0192	0,971	0,0235
Toxicity	0,731	0,0290	0,771	0,0314	0,720	0,0489	0,754	0,0531	**0,823**	**0,0334**	0,771	0,0180	0,783	0,0142	0,788	0,0142
average	0,9426		0,9406		0,9449		0,9477		0,9214		0,9098		0,9156		0,9228	
rank first	5		0		3		1		1		0		0		0	

TABLE VII
AVERAGE FITNESS VALUE OF THE PROPOSED FS METHOD USING EIGHT DIFFERENT TRANSFER FUNCTIONS AND STANDARD BINARIZATION RULE

	BKMA-SS1		BKMA-SS2		BKMA-SS3		BKMA-SS4		BKMA-SV1		BKMA-SV2		BKMA-SV3		BKMA-SV4	
	Avg Fitness	Std Fitness	Avg Fitness	Std Fitness	Avg Fitness	Std Fitness	Avg Fitness	Std Fitness	Avg Fitness	Std Fitness	Avg Fitness	Std Fitness	Avg Fitness	Std Fitness	Avg Fitness	Std Fitness
Breastcancer	**0,00954**	**0,002472**	0,02499	0,010374	0,01924	0,011920	0,01846	0,008977	0,02044	0,010289	0,02382	0,003313	0,02936	0,018579	0,02381	0,014998
BreastEW	0,04380	0,004254	0,03357	0,013639	**0,02419**	**0,010180**	0,03502	0,012051	0,04618	0,007825	0,04724	0,015609	0,03595	0,016740	0,05178	0,012156
Exactly	0,02518	0,003131	**0,01414**	**0,007467**	0,03394	0,039655	0,08010	0,046884	0,19632	0,026246	0,18429	0,021502	0,13844	0,057391	0,18215	0,088959
HeartEW	0,11363	0,011373	0,13184	0,033480	**0,10630**	**0,018796**	0,11363	0,026452	0,14608	0,050908	0,16099	0,027216	0,13153	0,015227	0,16522	0,014198
Ionosphere	0,07100	0,032395	0,09250	0,027347	0,07061	0,013015	0,09855	0,033178	0,08728	0,020107	0,10005	0,024612	0,06076	0,023423	0,09700	0,011027
KrVsKpEW	0,03474	0,004103	0,03507	0,004929	0,03241	0,003870	0,03370	0,002830	0,03696	0,003130	0,03949	0,005293	0,03891	0,008001	0,03767	0,011295
PenglungEW	0,11025	0,052593	0,08361	0,049071	0,11006	0,067054	0,07083	0,000243	0,05643	0,064407	0,04324	0,052667	0,04271	0,052660	0,09603	0,052388
Sonar	0,07974	0,017365	0,09466	0,040404	0,09848	0,025705	0,08442	0,048419	0,11374	0,038486	0,12271	0,026872	0,11220	0,044201	0,10214	0,018528
Vote	**0,00712**	**0,005826**	0,02305	0,019647	0,03290	0,018566	0,01627	0,012422	0,04577	0,022387	0,04385	0,022471	0,03612	0,008931	0,04589	0,012060
Wine	0,01478	0,013734	0,02079	0,021588	0,01401	0,013708	0,01463	0,021969	0,04862	0,051538	0,03645	0,026805	0,04743	0,013899	0,03126	0,017070
Zoo	**0,00425**	**0,000919**	0,01313	0,018447	0,03214	0,023182	0,02326	0,022873	0,03456	0,037359	0,02419	0,036211	0,04332	0,034375	0,02426	0,022570
Toxicity	0,29915	0,038502	0,28772	0,047249	0,25379	0,011058	0,28204	0,054719	0,24342	0,028841	0,22084	0,021173	0,21010	0,027998	0,22089	0,021144
average	0,0678		0,0713		0,0690		0,0726		0,0897		0,0873		0,0772		0,0898	
rank first	3		1		2		0		0		0		0		0	

TABLE VIII
AVERAGE FINESS VALUE OF THE PROPOSED FS METHOD USING EIGHT DIFFERENT TRANSFER FUNCTIONS AND ELITIST BINARIZATION RULE

	BKMA-ES1		BKMA-ES2		BKMA-ES3		BKMA-ES4		BKMA-EV1		BKMA-EV2		BKMA-EV3		BKMA-EV4	
	Avg Fitness	Std Fitness	Avg Fitness	Std Fitness	Avg Fitness	Std Fitness	Avg Fitness	Std Fitness	Avg Fitness	Std Fitness	Avg Fitness	Std Fitness	Avg Fitness	Std Fitness	Avg Fitness	Std Fitness
Breastcancer	0,02693	0,009368	0,01862	0,005099	0,02375	0,007792	0,02118	0,011707	0,03524	0,011797	0,03268	0,010745	0,03545	0,013153	0,02258	0,003831
BreastEW	0,03901	0,013881	0,02983	0,003590	0,03004	0,013219	0,03353	0,015306	0,04541	0,015615	0,05590	0,005509	0,05102	0,016410	0,04403	0,007533
Exactly	0,08826	0,059574	0,11255	0,080506	0,05414	0,041599	0,14514	0,119322	0,19245	0,021937	0,18452	0,046451	0,17401	0,024968	0,22077	0,034743
HeartEW	0,12856	0,025800	0,12824	0,031088	0,11659	0,020373	0,10942	0,031317	0,16081	0,015347	0,15007	0,038692	0,13531	0,043817	0,15007	0,041178
Ionosphere	0,05490	0,018414	**0,04870**	**0,014391**	0,06016	0,020946	0,05161	0,024401	0,07007	0,014636	0,09863	0,027139	0,08657	0,015450	0,08042	0,007304
KrVsKpEW	**0,03081**	**0,003250**	0,03943	0,004831	0,03855	0,008193	0,04312	0,008173	0,03508	0,004568	0,03638	0,004132	0,04254	0,003331	0,03817	0,002803
PenglungEW	0,06844	0,041725	0,05532	0,026751	0,04205	0,052864	**0,02868**	**0,032422**	0,10926	0,067138	0,09586	0,079073	0,06869	0,041692	0,02944	0,032523
Sonar	0,06412	0,011060	0,07326	0,020698	0,07402	0,020137	**0,06372**	**0,023314**	0,10365	0,022336	0,12669	0,017808	0,12323	0,038030	0,09372	0,017461
Vote	0,01636	0,019220	0,02321	0,006026	0,02881	0,016416	0,01245	0,007954	0,05511	0,030734	0,05071	0,014515	0,03859	0,016243	0,03218	0,013011
Wine	0,00813	0,010952	0,03045	0,024795	**0,00213**	**0,000887**	0,02074	0,013610	0,05411	0,036233	0,03229	0,001600	0,03798	0,010416	0,02584	0,026380
Zoo	0,03235	0,023650	0,03210	0,023439	0,01370	0,018491	0,01371	0,018465	0,04344	0,046130	0,03487	0,036003	0,04346	0,019282	0,03359	0,023156
Toxicity	0,26844	0,028741	0,26892	0,028411	0,27998	0,048335	0,25135	0,048654	**0,17556**	**0,033070**	0,22731	0,016707	0,21602	0,013252	0,20958	0,014068
average	0,0689		0,0717		0,0637		0,0662		0,0900		0,0938		0,0877		0,0817	
rank first	1		1		1		2		1		0		0		0	

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